



TRACK 3 • SMART HEALTH & SUPPLY CHAIN RESILIENCE

AUSHADH GRID

Federated AI for PHC medicine supply chains. It spots outbreaks early, forecasts demand from messy real-world data, and moves stock between centres before anyone runs out or throws medicine away.

SHORTAGES, WASTE AND BAD DATA AT THE SAME TIME, IN THE SAME DISTRICT.

Stock-outs next to surplus

One PHC turns patients away while another 40 km away holds months of the same medicine. Nobody can see both.

Medicine expires on the shelf

Quiet centres hold lots that expire unused, while busy centres nearby are short of the same item.

Bad data, bad forecasts

A PHC that ran out records zero use, so models learn that demand is low. Late reports and typos add to the error.

Outbreaks declared late

Fever and diarrhoea visits rise days before confirmation, while ORS and IV fluid demand triples.

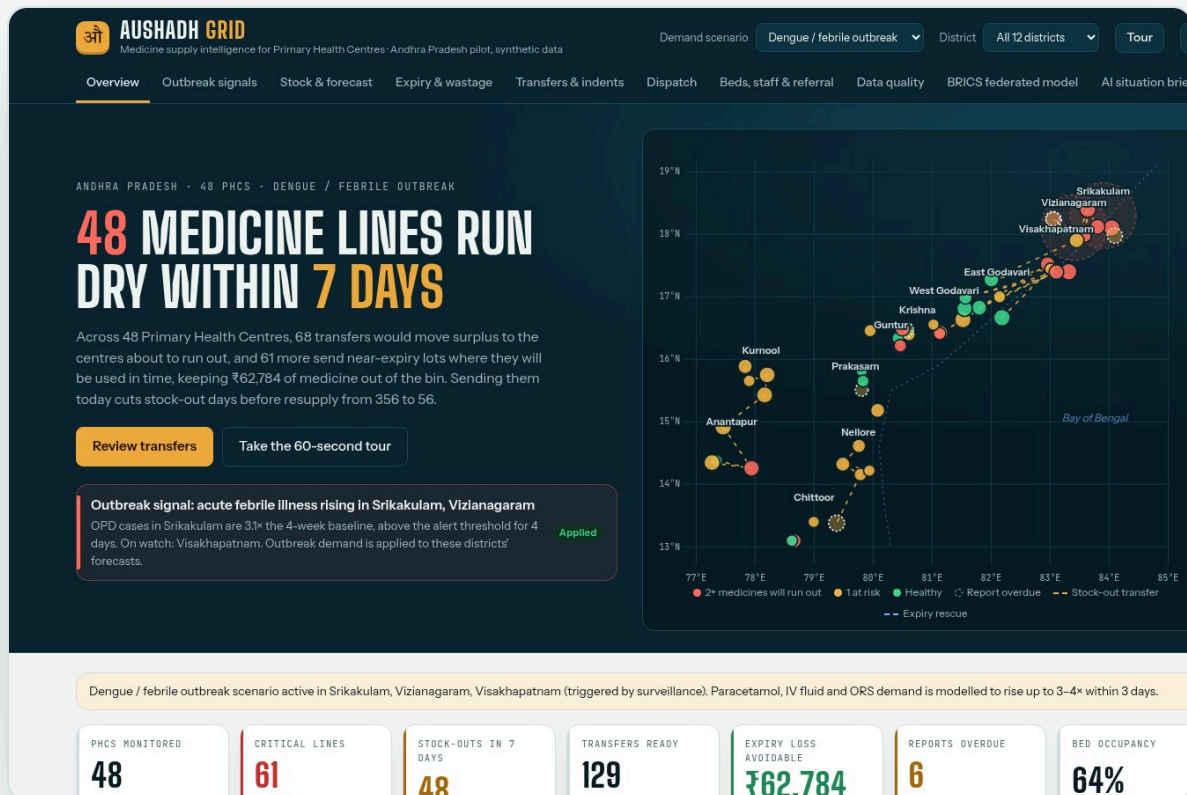
Cold chain breaks

Insulin, vaccines and oxytocin need 2–8 °C. A faulty fridge makes restocking pointless.

Tight budgets

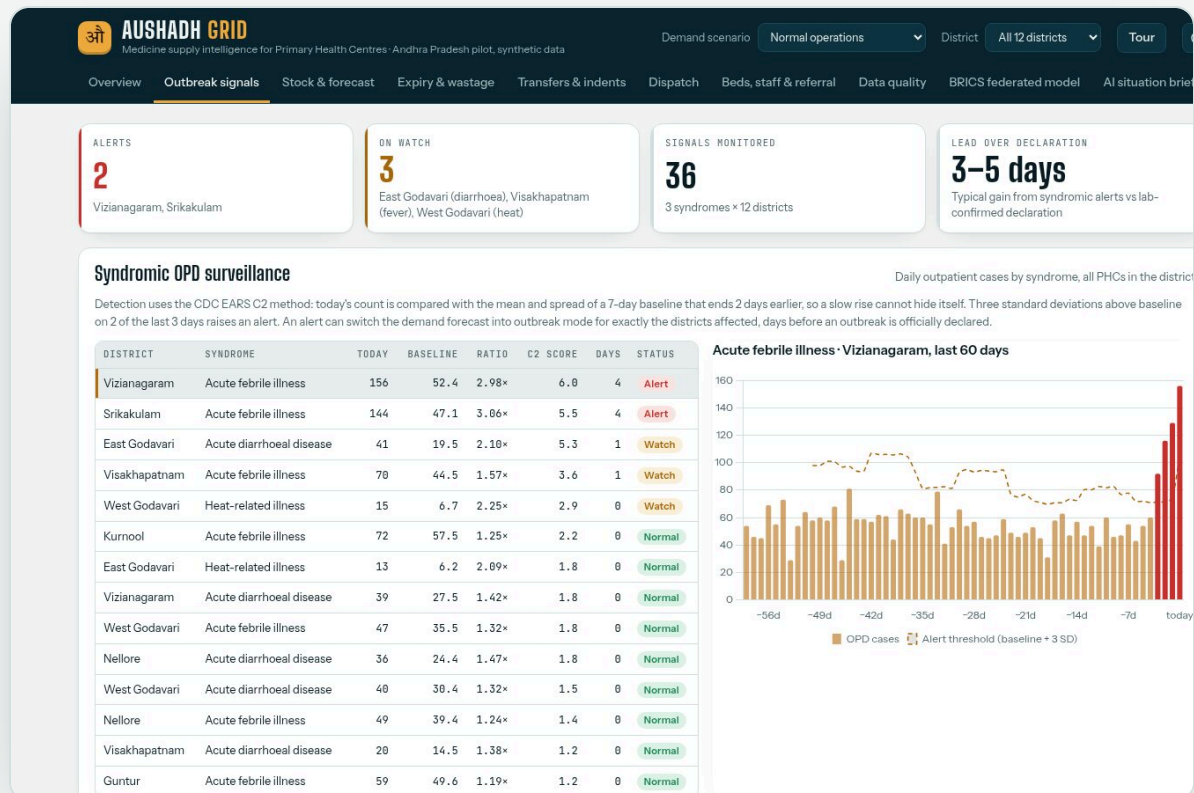
Indents must put vital medicines first, following WHO VEN practice, with cost visible.

ONE CONSOLE FOR EVERY PHC: SIGNALS, STOCK, EXPIRY, TRANSFERS, BEDS AND STAFF.



- **Outbreak signals** from clinic visit counts feed straight into the demand forecast
- **Forecasts that survive messy data:** stock-out days, typos and late reports are handled
- **Expiry simulation** of every lot, first-expiry-first-out
- **Transfer orders** that prevent stock-outs and rescue expiring stock, cold-chain aware
- **SMS both ways:** stock reports in with typo checks, work orders out to every pharmacist
- **Proof and logistics:** stock-out days with vs without the plan, and transfers grouped into vehicle runs
- **Referral finder**, federated BRICS model and Gemini briefs in any language

CLINIC VISIT DATA TRIGGERS OUTBREAK DEMAND FOR EXACTLY THE AFFECTED DISTRICTS.



Daily fever, diarrhoea and heat-illness OPD counts per district are checked with the **CDC EARS C2** method. The baseline is 7 days ending 2 days back, and an alert needs 3 SD above it on 2 of the last 3 days.

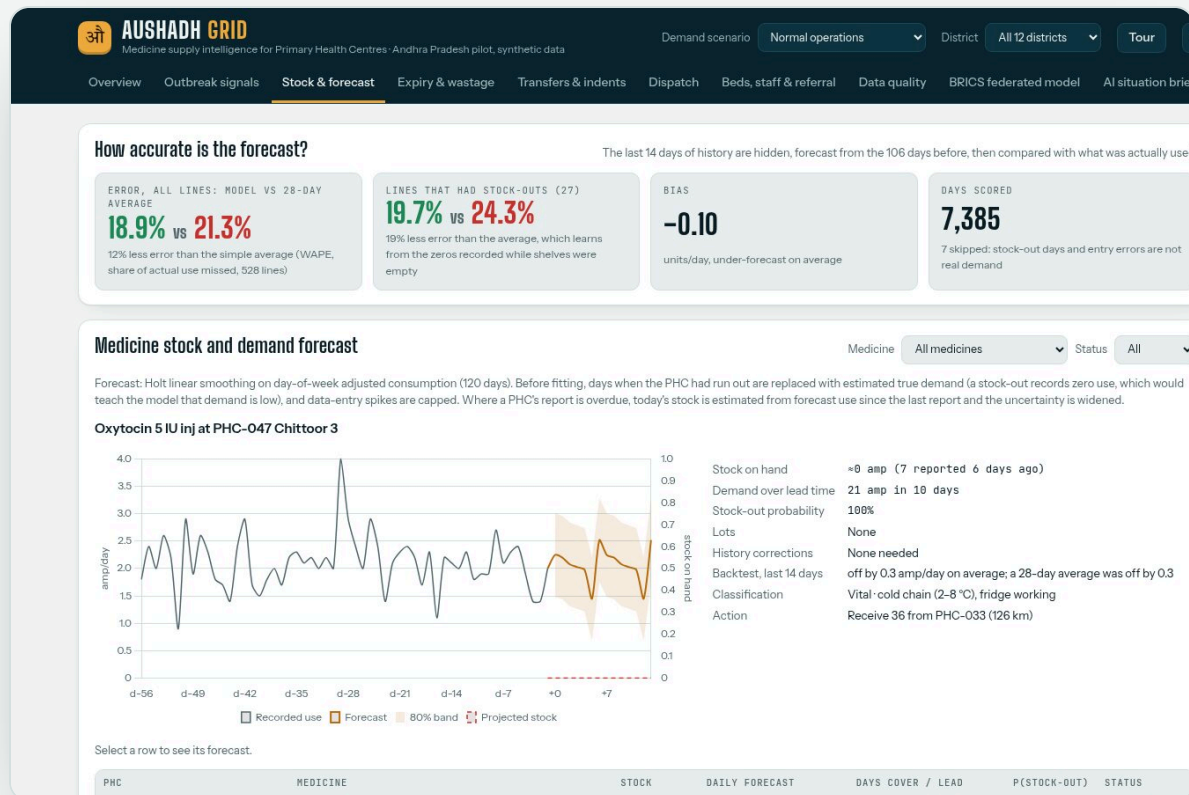
Pilot signal

3.1×

fever visits in Srikakulam vs the 4-week baseline, 4 days above threshold. Vizianagaram alerts too; Visakhapatnam is on watch.

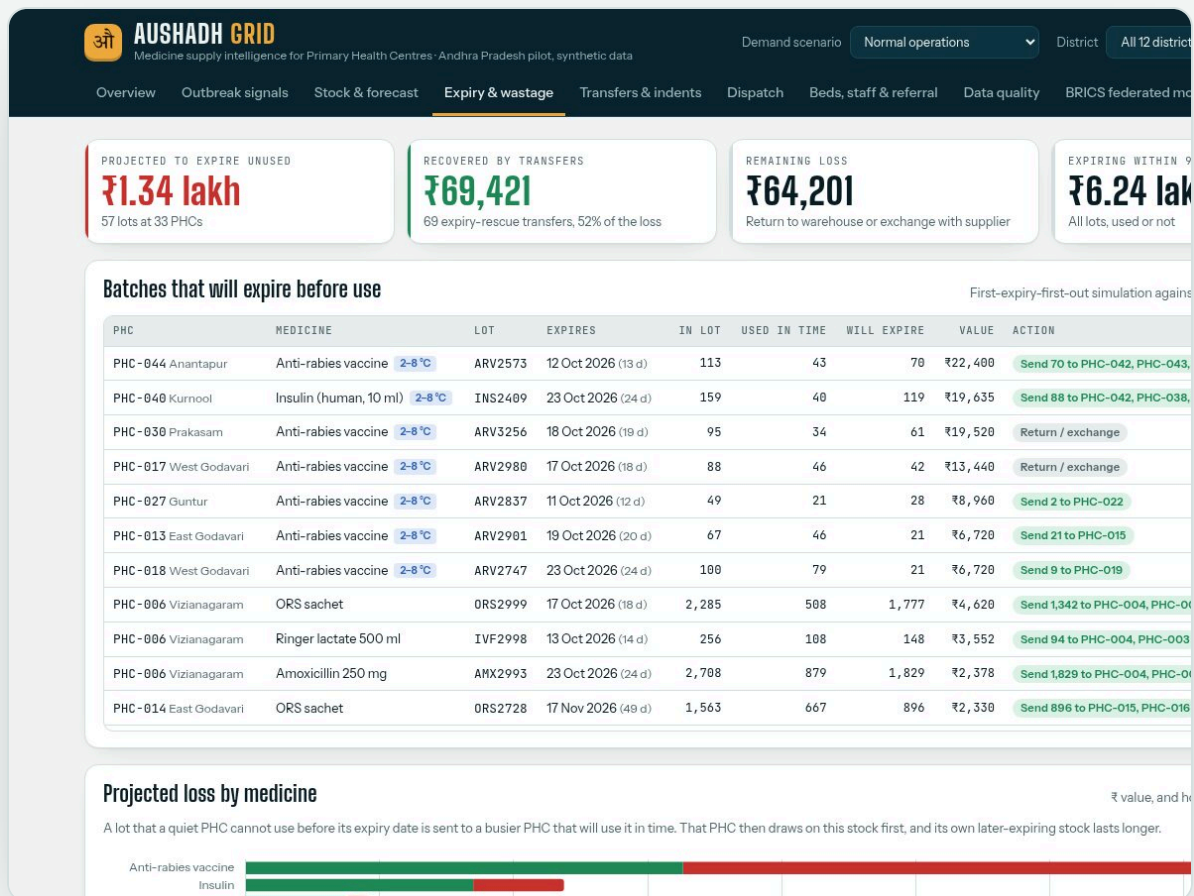
One click applies outbreak demand to those three districts. Critical medicine lines rise from 50 to 61, and transfers are re-planned at once.

STOCK-OUT DAYS ARE NOT ZERO DEMAND. THE MODEL KNOWS THE DIFFERENCE.



- **Censored demand:** 3+ zero days at a PHC that normally dispenses an item are re-estimated. The pilot restores **298 stock-out days**.
- **Entry errors:** days above 4× their local median are capped (**27** caught). SMS jumps of more than 60 days of demand are held for confirmation.
- **Late reports:** today's stock is estimated from forecast use, and uncertainty widens with report age.
- **Forecast:** exponential smoothing with weekday seasonality, tuned on an earlier window. On a 14-day holdout it misses **18.9%** of actual use vs 21.3% for a 28-day average, and 19.7% vs 24.3% on lines that recently ran out.

MOVE STOCK THAT ALREADY EXISTS, BEFORE IT RUNS OUT OR EXPIRES.



Stock-outs averted

44

68 transfer orders, 66 km
average haul

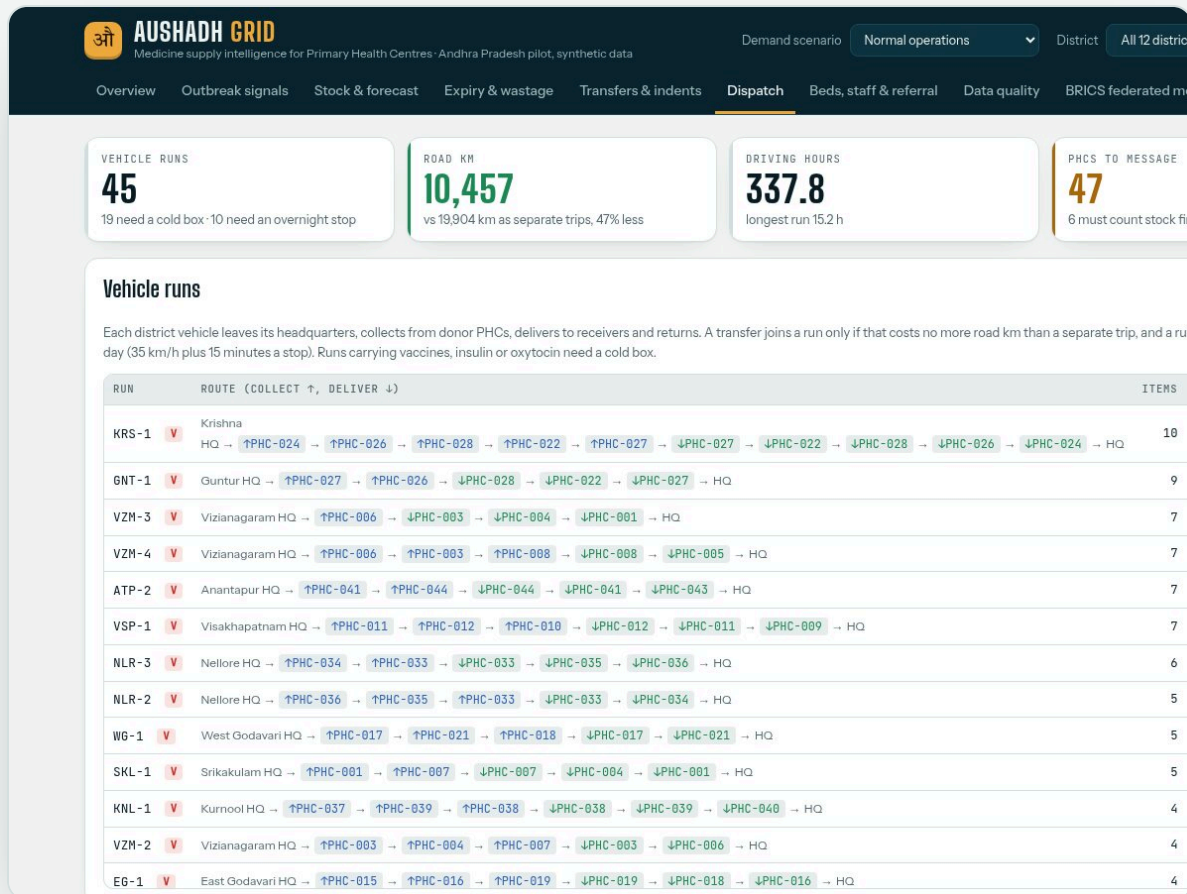
Expiring stock rescued

50%

₹62,784 of ₹1.26 lakh
projected loss

- Vital medicines first; donors keep their own safety stock
- Cold-chain items only between working fridges, within 150 km
- 20 warehouse indents (12 vital, ₹46,349), flagged when a fridge must be repaired first

PROOF THE PLAN WORKS, AND THE RUN SHEETS AND SMS TO CARRY IT OUT.



Vital-medicine stock-out days

179 → 16

before routine resupply, no action vs the plan

Road km

-47%

45 vehicle runs, 10,457 km vs 19,904 km as separate trips

- Day-by-day simulation of all 528 lines; unmet demand counted as lost, not delayed
- Runs collect then deliver, fit an 8-hour day, and flag a cold box
- One SMS per PHC in the codes pharmacists already use; 6 with stale reports count stock first

FROM A CLINIC REGISTER AND A PHARMACIST'S SMS TO AN APPROVED ORDER.

Detect

Surveillance

EARS C2 on syndromic OPD counts sets outbreak multipliers per district and medicine.

Clean

Data repair

Censored demand, typo capping, stale-report estimation, SMS plausibility checks.

Predict

Forecast & risk

Backtested exponential smoothing, stock-out probability, FEFO expiry simulation per lot.

Act

Optimise

Cold-chain-aware transfers, expiry rescue, VEN-ranked indents, vehicle runs, SMS work orders.

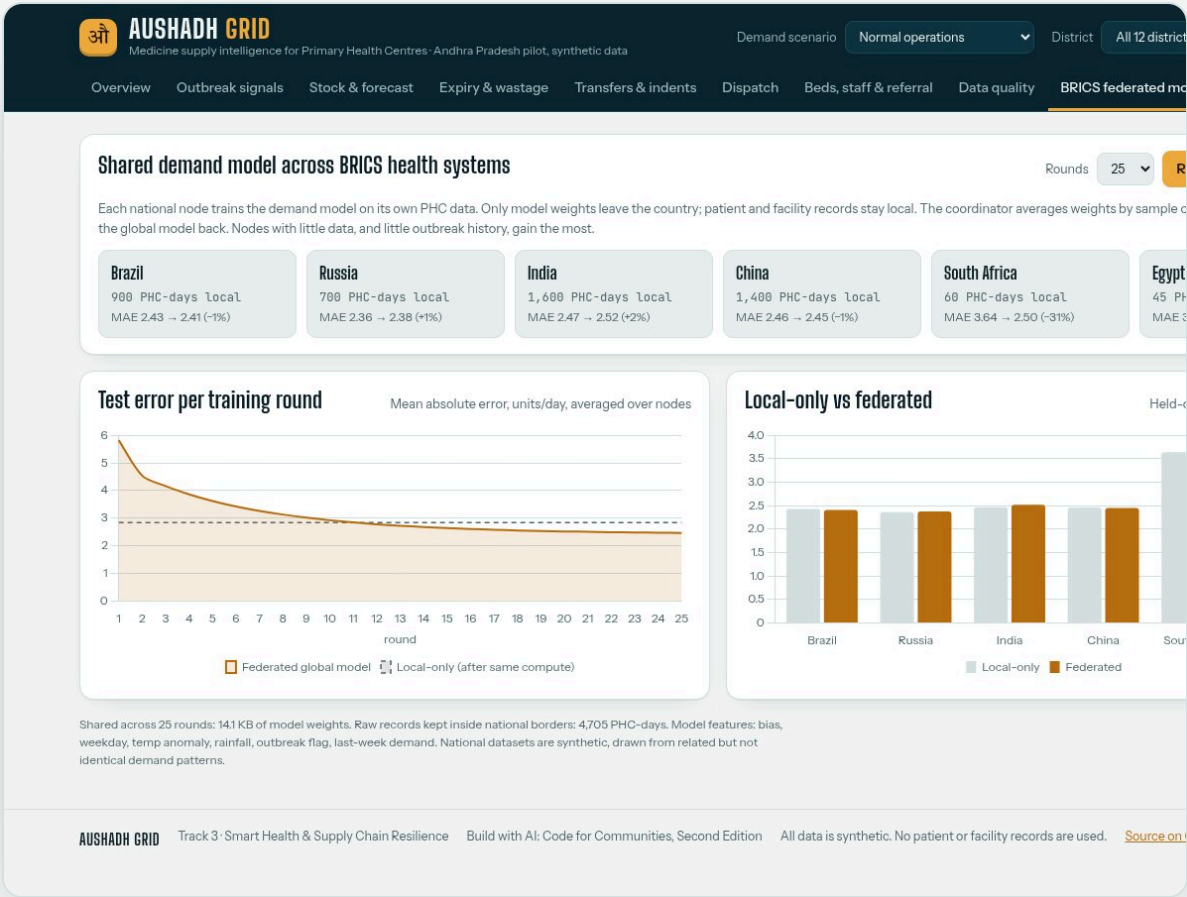
Learn together

A federated demand model is shared across BRICS health systems. Only weights cross borders.

Explain

Gemini writes a situation brief from the exact dashboard state, in Hindi, Telugu, Portuguese, Russian or Chinese. The input JSON is shown on screen.

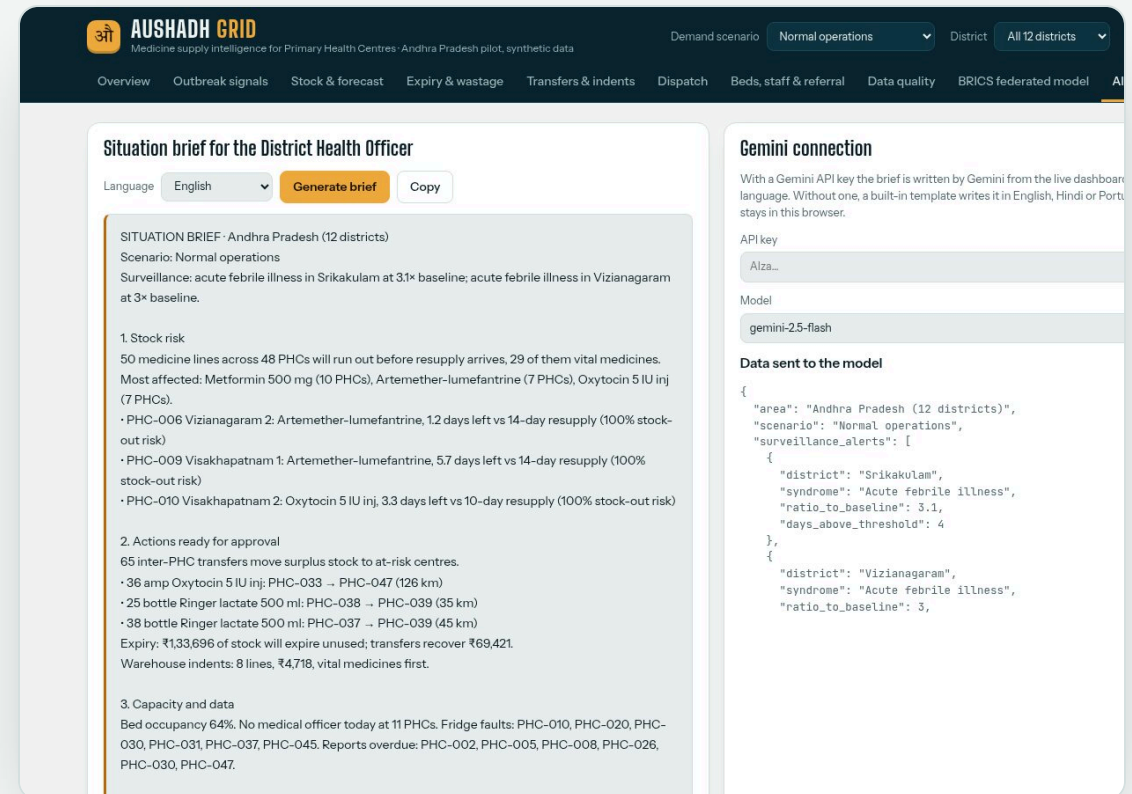
COUNTRIES SHARE WHAT THE MODEL LEARNED, NEVER THE PATIENT DATA.



Each national node trains on its own PHC records. A coordinator averages the weights (FedAvg) and returns a global model.

NODE	LOCAL DATA	ERROR CHANGE
South Africa	60 PHC-days	-31%
Egypt	45 PHC-days	-31%
India, China, Brazil, Russia	700-1,600	within ±2%

Small systems that have rarely seen an outbreak learn outbreak demand from their partners. 14 KB of weights are shared and 0 raw records move.



OPEN, OFFLINE-FIRST, AND READY TO PLUG INTO REAL SYSTEMS.

Prototype today

- Runs in any browser with no install or backend
- Open source (MIT) and multilingual
- 48 PHCs, 12 districts, 528 medicine lines, 36 surveillance signals, 6 federated nodes
- CSV import so a district can try it on its own ledger
- Gemini integration with offline fallback

Production path

- Cloud Run API, AlloyDB / Firestore lot ledger, BigQuery history
- Vertex AI forecasting and Gemini
- Pub/Sub webhooks for SMS, WhatsApp and IVR
- Flower or TensorFlow Federated for cross-border training
- Integrations: state drug systems (DVDMS / eAushadhi), IDSP/IHIP surveillance, ABDM facility registry

Impact: fewer patients turned away, less medicine thrown away, outbreak stock in place days earlier, and a shared model that makes every partner's forecasts better.